GreenTwin

A calibrated physics digital twin for every UK home: Green Home Score, retrofit path, per-fuel savings, carbon and value, each with Bayesian confidence, and green home lending for Lloyds

Megathon X: Singularity · Problem Statement 2 (Lloyds): Green Home Score and Retrofit Recommendation Engine

Team MegaMind · Milestone 1 submission: approach and idea · 10 October 2026

In one sentence. We model each home's physics, calibrate it with Bayesian inference against real metered energy, and use that one documented model to compute every number the customer sees. The LLM only explains those numbers. A 3D twin lets the customer add any retrofit and watch the effect, and a Lloyds dashboard turns this into targeted green-lending demand.

1. Executive summary

Customers want greener homes but do not know where to start; Lloyds wants to lend for retrofits but needs to know who will benefit. GreenTwin answers the brief's four questions (how green is my home, what should I improve, how much energy and carbon could I save, could it raise my value) with documented models, and adds what a bank needs to act.

- **One source of truth.** A Level-1 RC (resistance-capacitance) physics twin of each home, calibrated with Bayesian inference against metered gas and electricity, produces the score inputs, per-fuel savings, carbon, improvement potential and the inputs to the value model. Retrofits are parameter edits, so interactions (insulation before a heat pump) are computed, never summed.
- **Bayesian confidence everywhere.** Every number has a 90% credible interval; every retrofit has P(suitable) = P(feasible) × P(net benefit); every package on the Pareto front has P(on front).
- **Explainable score.** Fixed weights from the brief with monotonic, documented sub-score mappings and a contribution waterfall; an EBM (glass-box) model explains learned effects, with XGBoost as baseline.
- **Grounded GenAI.** A tool-only LLM that cannot calculate: what-ifs rerun the models, and a numeric validator rejects any number not in the computed output. Jev, a fast System-1 model, routes questions and screens feasibility with calibrated confidence; rule vetoes always win.
- **Value without overclaiming.** Real EPC-linked sale prices; a hedonic quantile model, a green-twin matched-pair study and a doubly robust (DR) learner; always a range, always labelled indicative.
- **Business plan.** A free customer-acquisition app plus agent visits that build the twin on site; a Lloyds dashboard with a locality heat map, segments and a Green Profits view of lending, retention and acquisition.
- **Experience.** A Three.js twin of the house, outside and inside, where every retrofit appears on the model and the computed numbers update, plus a staged retrofit timeline.

How the design maps to the judging criteria

Criterion (weight)	What GreenTwin contributes
ML and recommendation quality (20%)	Real public data + physics-generated labels validated against NEED; property-level splits; baselines for every model; feasible, ordered recommendations
Green Home Score and insight (15%)	Fixed weights, monotonic documented mappings, contribution waterfall, score interval and band probabilities
Savings, carbon and value (15%)	Per-fuel savings from the calibrated twin with interactions; documented emission factors; value range from three methods with causal caveats
GenAI and LLM (15%)	Tool-only LLM, rerun-on-what-if, numeric validator, 30–100 adversarial scenarios against the ≥90% / <5% targets
Business relevance (10%)	Green-loan journey, agent and self-serve channels, Lloyds heat map, segments, Green Profits
Architecture (10%)	One contract layer, cached deterministic runs, seeds and data versions, offline demo
Explainability and Responsible AI (5%)	Glass-box score and EBM, provenance on every number, privacy by design, human adviser for credit
UX (5%) and innovation (5%)	Interactive 3D twin inside and outside; customer memory; Jev System-1/System-2 routing

2. Business plan: Lloyds green home lending

GreenTwin is a free customer-acquisition and green-lending tool. It is built on sections 2, 3, 4, 7 and 10 of the problem statement and turns each into a step in a lending conversation.

Brief section	Role in the business
2 · Green Home Score	Shareable headline score that starts every conversation
3 · Poorly performing homes	Finds homes and areas with the most improvement potential, i.e. loan demand
4 · Retrofit recommendations	The prioritised feasible package becomes the loan's purpose and amount
7 · Property value	Indicative value range strengthens the case for borrowing
10 · Customer memory	Keeps each customer's plan, constraints and responses so follow-ups improve

Two channels

- **Agent-assisted.** A Lloyds agent visits, photographs the floor plan and rooms, and the app builds the twin on the spot (v1: photos plus manual element tags confirmed by the agent; v2: floor-plan parsing and element tagging with confidence scores). The customer sees the score, the retrofit path on the 3D house and what a green loan would mean.
- **Self-serve.** The customer looks up their EPC by postcode, uploads a home profile and explores improvements; a "talk to an adviser" button turns interest into a lead.

Customer journey

Score and key issues → recommended retrofit path on the 3D house → package cost minus grants = indicative loan amount → monthly repayment vs monthly bill saving ("cash-flow positive?") → value range → save the plan or book an adviser. Loan figures are illustrative, not an offer or advice; any credit decision follows Lloyds' regulated process with a human adviser.

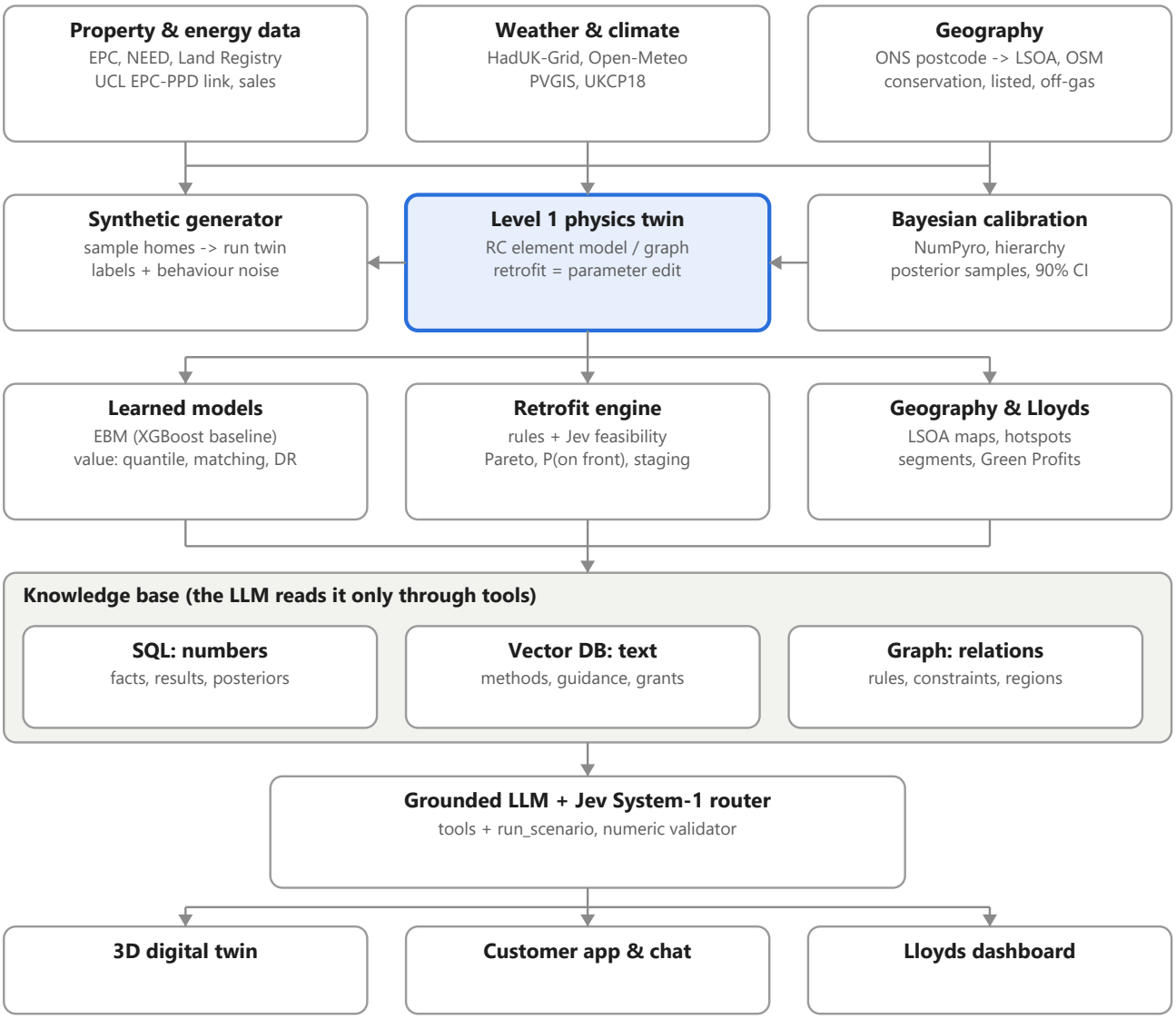
Lloyds dashboard

- **Locality heat map (centrepiece)** by LSOA or postcode district with switchable layers: share of green homes (score ≥ 65), share of Poor/Very poor homes, customers interested in going green (what-ifs run, plans saved, adviser requests), estimated retrofit finance demand, carbon reduction potential, heat-pump grant eligibility (off-gas-grid) and density of existing customers.
- **Segments**, e.g. "1930s semi, gas boiler, owner-occupier, low loan-to-value", with size, average loan need, take-up probability from an uplift model and carbon potential, ranked for outreach.
- **Green Profits view**: profit from green lending (balances × net interest margin, plus fees), retention (lower churn among customers who retrofit, valued as customer lifetime value) and acquisition (app users who become customers, net of acquisition and visit costs), with financed-emissions reduction alongside. Every assumption is a slider; results are Monte Carlo ranges.
- **Funnel**: app users → home profiles → saved plans → visits → applications → loans drawn.

The brief forbids real customer data, so the dashboard runs on a synthetic Lloyds customer base laid over real geography and real EPC distributions, clearly labelled. The idea that green homes carry lower credit risk is presented as a hypothesis, not a finding.

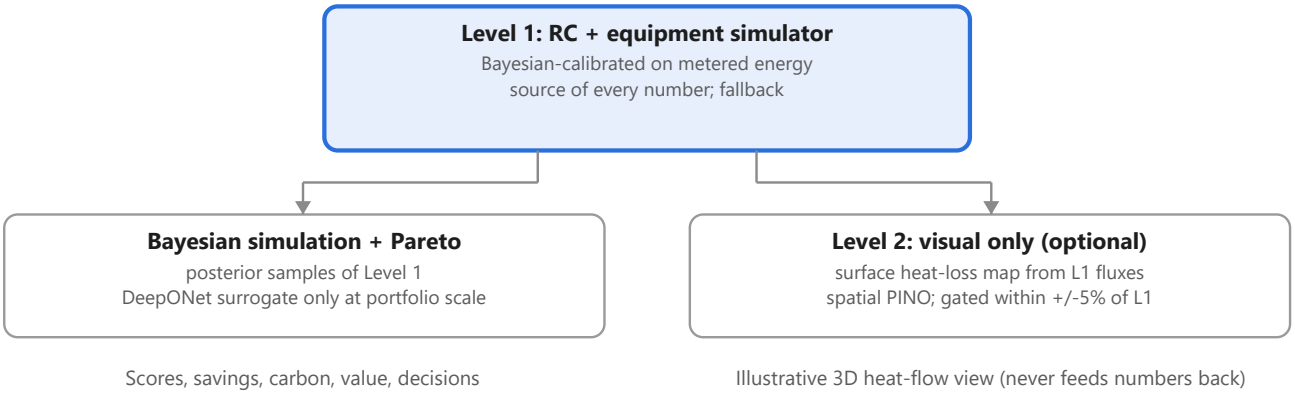
3. Solution architecture

Data and environment feed one physics twin; calibration, learned models and analysis build on it; all results land in a knowledge base that the LLM can reach only through tools.



A what-if from the 3D twin or the chat becomes a **ScenarioSpec** (retrofits, sizes, weather, samples). **run_scenario** edits the twin's parameters, propagates the posterior samples and the downstream models, and returns a **ScenarioResult** with provenance, which the UI renders and the LLM explains. Every module exchanges the same versioned data contracts (Pydantic models with a TypeScript mirror).

Two-level simulation



Level 1 produces every number and drives every decision. Level 2 only visualises, and only when its integrated heat flow agrees with Level 1; information never flows back.

4. Green Home Score

$S = \Sigma\, w_k\, s_k$ with the brief's fixed weights; each sub-score s_k is 0–100 and monotonic (a better state never gives a lower sub-score, enforced by unit tests). The final score is rounded before classification.

Component (weight)	Input	Mapping (summary)
Energy efficiency (25%)	EPC SAP score / band	SAP 1–100 directly; band midpoints A 96 … G 10 if only the band is known
Energy intensity (20%)	Weather-normalised metered purchased energy per m^2	Piecewise linear: 50 kWh/m^2 → 100, 150 → 70, 250 → 40, 400 → 0
Insulation (15%)	Roof, walls, floor	Roof 40% / walls 45% / floor 15%, re-normalised for flats; loft depth 0–270 mm → 0–100; solid uninsulated 0 … as-built 100
Heating system (15%)	Main system and age	Ground/air-source heat pump 100/95 … condensing gas 55 … oil 20, solid fuel 5; old boilers lose points
Windows and glazing (10%)	Glazing types by share	Single 0, secondary 50, double pre-2002 60, post-2002 80, triple 100
Renewable energy (10%)	PV kWp, battery, solar thermal	0–5 kWp → 0–100, plus small bonuses, capped
Smart energy management (5%)	Controls present	None 0 → programmer + thermostat 40 → smart + TRVs 90 → smart + zoning + smart meter 100

Double counting. Energy efficiency reflects the recorded rating (a modelled overall outcome) and energy intensity reflects metered use per m^2, as the brief requires. Insulation, heating, glazing, renewables and controls are the levers a customer can change, so their overlap with the rating is accepted and documented. Using kWh/m^2 (weather-normalised, with peer context) stops large homes being classed as inefficient for size alone.

Classification: 80–100 Highly efficient · 65–79 Good · 50–64 Moderate · 30–49 Poor · 0–29 Very poor. With Bayesian inputs we also report the score interval and band probabilities, e.g. 38 (90% CI 33–44), P(Poor) = 0.81. Poor and Very poor homes are flagged with their key issues and ranked by improvement potential $100 \times (E_{baseline} - E_{after}) / E_{baseline}$ on purchased energy, reported per fuel; if the baseline is zero or missing the metric is shown as unavailable with the reason.

5. Physics digital twin and synthetic data

- **Level 1 twin (v1).** Steady-state RC element model: heat loss = Σ U·A + ventilation, driven by regional degree days; space heat and hot water converted to fuels by boiler efficiency or heat-pump SCOP; PV yield from PVGIS reduces grid import. Outputs per-fuel kWh and heat loss per envelope element (which colours the 3D twin).
- **v2 (stretch, GPU).** A differentiable RC graph (4–6 thermal nodes) whose conductances are predicted from EPC element features and solved as an ODE; energy conservation and monotonicity hold by construction.
- **Synthetic generator.** Samples homes from EPC / English Housing Survey joint distributions, attaches real weather, runs the twin and adds behaviour noise from NEED-calibrated priors. It produces monthly usage, per-retrofit reference labels and (for synthetic homes) property values from a hedonic model fitted on real sales plus an explicit, documented green-premium term. All rules are disclosed; agreement with synthetic targets is not claimed as real-world accuracy, and the generator is validated against real NEED distributions.
- **Extrapolation** (new retrofit combinations, future climate) is done only by the physics twin; an out-of-distribution flag warns when a home is far from the training data.

6. Bayesian confidence in every output

NumPyro calibrates each home's heat-loss coefficient, heating setpoint, baseload and heat-pump efficiency against its metered monthly energy, with hierarchical priors by property type × region (RdSAP default U-values, Electrification of Heat trial SCOP). Missing months are latent, never zero. Fits are offline; per-home updates use variational inference in under a second.

Output	Shown as	Computed from
Green Home Score	38 (90% CI 33–44); P(Poor) = 0.81	Deterministic formula on posterior samples of uncertain inputs
Per-fuel savings	Gas +6,100 kWh/yr (CI …); electricity −1,900 kWh/yr	Twin rerun on each posterior sample, before vs after
Carbon	tCO_2e/yr with CI	Per-fuel savings × documented emission factors
Value impact	+2–5% range, indicative	Posterior predictive of the value model, before vs after
Retrofit suitability	High / Medium / Low	P(suitable) = P(feasible) × P(net benefit): ≥ 0.8 High, 0.5–0.8 Medium
Data confidence	High / Medium / Low	Months of meter data, EPC age, imputed fields; drives interval width

Illustrative figures only. Calibration is checked on held-out homes: 90% intervals should contain the truth about 90% of the time (coverage, CRPS, PIT histogram, R-hat).

7. Retrofit engine

- **Feasibility, layer 1 (rules).** Deterministic vetoes with reason codes: no loft, not a cavity wall, external insulation in a conservation area or on a listed building, no roof access for PV, north-facing roof, no outdoor space for a heat-pump unit, already installed. A rule veto cannot be overridden.
- **Feasibility, layer 2 (Jev).** For retrofits that pass the rules, Jev reads home facts plus free text (EPC descriptions, agent notes, planning details) and returns feasible / with conditions / needs survey / infeasible with a calibrated confidence; low confidence becomes "needs survey", never a silent pass.
- **Ordering.** Fabric first, then glazing, controls, heating system (sized on reduced demand), then generation and storage. Marginal savings are computed in sequence on the twin, so interactions are explicit.
- **Multi-objective Pareto.** All feasible packages (≤ 256) are enumerated and evaluated on net cost after grants, carbon, bill saving, score gain and value. The front is computed for every posterior sample, giving **P(on front)**; the knee point is the default suggestion and a budget slider shrinks the front live.

8. Energy savings and carbon

Savings are estimated per retrofit (marginal, in sequence) and for the package, separately for electricity and gas, against one consistent baseline. An increase is a negative saving: a heat pump removes gas use and adds electricity. Carbon is $\Delta C = \Sigma_f \Delta E_f \times EF_f$ with DESNZ 2025 conversion factors (electricity about 0.177 $kgCO_2e$/kWh; gas on a gross calorific basis to match meter readings; both to be confirmed against the official tables, with units and source recorded). All savings are labelled estimates, not guarantees.

9. Property value impact

Value impact is always a range and always labelled an indicative estimate based on historical and market relationships in our data, not a proven causal effect.

- **Data.** The UCL attribute-linked Land Registry Price Paid and EPC dataset (millions of sales with floor area, rooms and energy information) plus the UK House Price Index for market conditions. The current valuation used as baseline is kept separate from historical sale prices.
- **Hedonic quantile model.** Price quantiles from location, size, type, age, period and EPC band; predict the same home before and after the proposed improvements, holding everything else constant.
- **Green-twin matching.** Pairs of real homes that match on type, built form, age band, tenure, LSOA, sale quarter and floor area (±10%) but differ by at least 15 score points; balance checked (standardised mean difference < 0.1). Measures the price and running-cost difference per 10 points, benchmarked against published estimates (e.g. Nationwide: A/B about 1.6–1.7% above comparable D homes).
- **DR-learner.** On repeat sales, a doubly robust causal meta-learner estimates how much an EPC upgrade changes price for different homes, with sensitivity analysis and refutation tests for unmeasured confounding (such as other renovations).

10. Grounded LLM and what-if questions

- **Tool-only.** The LLM can call get_home_facts, get_results, run_scenario, search_docs and templated graph queries. It never computes; every what-if reruns run_scenario first.
- **Validator.** Every number in an answer is extracted and matched against that turn's tool outputs; a mismatch triggers regeneration, then a template fallback. A second Jev check asks whether every claim is supported by the JSON.
- **System 1 / System 2 routing.** Jev classifies the question (numeric, comparative, feasibility, out-of-scope) and fills the scenario with a calibrated confidence; below the threshold the LLM asks a clarifying question. Numbers always come from the physics tool (System 2). Without a Jev key, Claude with structured output fills the same interface.
- **Honesty.** Missing or conflicting data, infeasible retrofits and electricity increases are stated plainly; loan figures are illustrative; a human adviser makes credit decisions.

11. Weather, climate and geography

- **Weather normalisation.** Per-home degree-day regression (kWh = baseload + UA_{eff} × HDD, base 15.5 °C) separates a cold year from a poor house and annualises missing months from that month's actual degree days.
- **Sources.** Met Office HadUK-Grid (1 km daily temperature, Open Government Licence) and UKCP18 for documented figures; Open-Meteo for hourly, forecast and CMIP6 projections; PVGIS for solar yield. Responses are cached.
- **Trends and lifetime value.** Regional heating- and cooling-degree-day trends (with intervals) on the map; future-climate scenarios turn first-year savings into lifetime savings and carbon; an overheating flag for homes where insulation without shading could raise summer temperatures.
- **Geography.** Postcode → LSOA via the ONS directory; conservation areas, listed buildings and off-gas-grid status feed feasibility and grants; hotspot analysis finds clusters of poor homes; the spatial hierarchy lets sparse areas borrow strength in the Bayesian model.

12. Models we build

Model	Kind	Purpose	Baseline
Green Home Score	Deterministic	Score, band, contributions	—
Level-1 twin + synthetic generator	Physics / simulator	Per-fuel energy, element heat loss, labels	NEED medians
Energy-intensity EBM	Glass-box ML	Expected kWh/m² per fuel; prior and residual	XGBoost; NEED median
Bayesian calibration	NumPyro hierarchical	Posterior parameters → intervals	LightGBM quantile; conformal
Improvement-potential ranking	Twin + ranker check	Rank poor homes by % reduction	Rank by total kWh
Retrofit relevance prior	EBM classifier	P(EPC would recommend a measure)	Rules only
Value: quantile / matching / DR-learner	ML + causal	Indicative value range	Hedonic OLS
Pareto optimiser	Optimisation	Front, P(on front), staging	Greedy kWh per £
Jev calibration	Isotonic	Calibrated routing / feasibility confidence	Claude structured output
Stretch (GPU)	Physics-ML / operators / CV	RC graph, DeepONet surrogate, GraphSAGE, spatial PINO, photo-to-twin	Level-1 twin

13. Data

Dataset	Use	Licence / access
EPC domestic certificates (England & Wales)	Property features, ratings, recommendations, feasibility	Free; GOV.UK One Login; bulk + API
NEED anonymised 50k	Real metered gas and electricity with installed measures	Open Government Licence
UCL Land Registry–EPC linked dataset	Sales with energy information for value models	UK Data Service ReShare
HM Land Registry Price Paid; UK HPI	Sale history; market conditions	OGL (attribution)
London smart meters (half-hourly)	Load shapes, solar self-consumption	Open (London Datastore)
Electrification of Heat trial	Real heat-pump efficiency priors	UK Data Service
English Housing Survey	Stock distributions for the generator	Official statistics
HadUK-Grid, UKCP18, Open-Meteo, PVGIS	Weather, climate, solar	OGL / CC BY 4.0 / free
ONS postcode directory, planning.data.gov.uk, Historic England, OSM	Geography and planning constraints	Open
DESNZ GHG factors, Ofgem price cap, BEIS retrofit costs	Carbon, bills, costs	Official publications

No real personal or customer data is used. Splits are by property (or customer), assigned once; EPC "potential" fields are excluded from model inputs as leakage. Missing months and fields are handled explicitly and documented. Three synthetic showcase homes drive the demo: a 1905 solid-wall terrace in a conservation area, a 1972 cavity semi, and an all-electric third-floor flat.

14. User experience

- **3D twin (Three.js).** The customer's house outside and inside (cutaway), with a thermal view coloured by Level-1 heat loss per element. Every retrofit can be added and appears on the model: insulation layers in the loft and walls, new windows, the heat-pump unit outside and cylinder inside, panels on the best roof face, a thermostat on the wall. Infeasible items are greyed out with the reason. A live panel shows score, per-fuel kWh, carbon, bills and value with intervals; a timeline plays the staged retrofit path year by year.
- **Customer app.** Home profile, score and contribution waterfall, retrofit plan, Pareto chart, green-loan view, "green twins of your home" (matched greener homes shown as ranges, never addresses), what-if chat and plan history.
- **Agent capture.** EPC pre-fill, floor-plan and room photos, element checklist, consent and blurring, twin on site.
- **Lloyds dashboard.** Heat map with switchable layers, segments, Green Profits and funnel.

15. Evaluation

Component	Held-out metric	Baseline
Energy model	MAE / MAPE per fuel; 90% interval coverage; CRPS	NEED median by type, size, region
Savings	Error vs held-out reference labels and NEED before/after cases	Sum of illustrative percentages
Calibration	Coverage ≈ 90%; flat PIT; R-hat < 1.01	Quantile regression; conformal
Improvement ranking	Spearman / NDCG	Rank by total consumption
Recommendations	Precision@k vs feasibility labels; zero infeasible recommended	Recommend everything
Value	Interval coverage on held-out sales; recovery of a known synthetic premium	Hedonic OLS
Score	Monotonicity sweep: 0 violations; reproducibility	—
LLM	Groundedness ≥ 90%, accuracy ≥ 90%, unsupported < 5% on 30–100 adversarial scenarios	LLM without validator
Pareto	Hypervolume	Greedy kWh per £
Jev	Routing accuracy, feasibility precision/recall, calibration error	Rules; Claude structured output

Adversarial scenarios cover missing months, conflicting data (rating A but 400 kWh/m^2), infeasible requests (a heat pump in a third-floor flat), electricity rising after a heat-pump switch and too little data to estimate value. Every reported metric is reproducible from a seeded script.

16. Responsible AI

- **Privacy.** Public and synthetic data only; location shown as postcode district or LSOA; interior photos need explicit consent, are blurred and kept only to build the twin.
- **Explainability.** Glass-box score and EBM; provenance on every number; reasons from rule codes.
- **Hallucination control.** Tool-only LLM, numeric validator, cached and replayable calls.
- **Bias.** Checks that large, older or electric homes are not penalised for size or fuel alone; per-segment metrics.
- **Human oversight.** Loans are illustrative; credit decisions stay with a human adviser; low-confidence feasibility becomes a survey, not an automatic answer.
- **Limits stated.** Synthetic labels, observational value effects and vendor model claims are labelled as such.

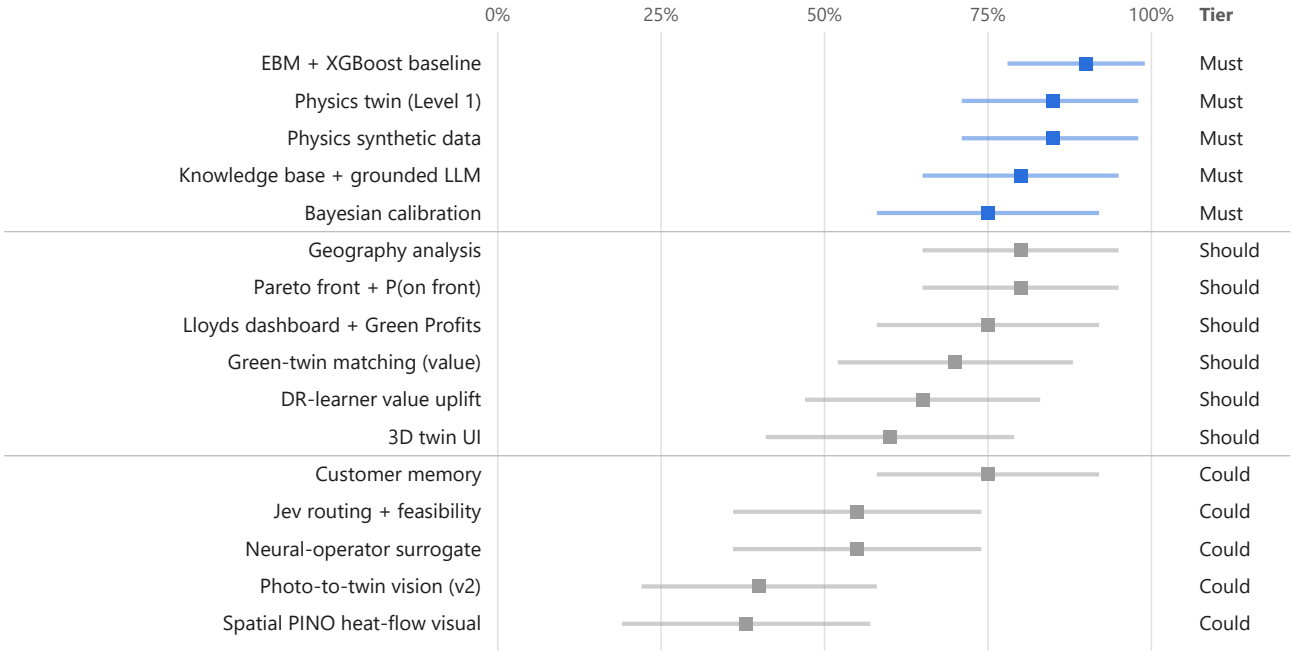
17. Build plan and team

Five members with clear ownership: Data & Geography; Physics & Bayesian; ML, Value & Causal; LLM, Knowledge base, Jev & API; Front-end & 3D. We build a thin end-to-end path first, then deepen in order of rubric weight × confidence; each checkpoint leaves a submittable system.

Tier	Contents	Checkpoint
A — required outputs	Contracts and splits; Level-1 twin; score; rules and catalogue; run_scenario; synthetic generator; EBM/XGBoost; value ranges	Score, recommendations, per-fuel savings, carbon and value for 3 homes
B — grounded product	Tool-only LLM + validator; 2D dashboard and chat; evaluation harness	Complete, evaluated submission
C — differentiators	Weather normalisation; Bayesian calibration; Pareto; geography and Lloyds dashboard; 3D twin; Jev; matching and DR-learner; memory and agent capture	Demo build; feature freeze
D — stretch (GPU)	Differentiable RC graph; DeepONet; GraphSAGE; spatial PINO; photo-to-twin; UKCP18 lifetime savings	Only after C

Confidence map

Each technique carries a Bayesian confidence score: a Beta prior on the chance of finishing it well (dot = mean, line ≈ 80% interval). These are planning estimates, updated at each checkpoint.



Main risks

Risk	Mitigation
Too many techniques	Tiers, checkpoints and stop rules; stretch work only after the demo build
Synthetic labels mirror the generator	Validate against NEED; state the limitation
Value read as causal	Ranges, "indicative", sensitivity and refutation tests
API keys or Jev access unavailable	Same interfaces with Claude, rules or cached fallbacks
Live demo failure	Precomputed showcase packages, offline mode, recorded video

References

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